

Simulator-Based Offline RL for Sequential Movie Recommendation

4

Policies Compared

500

Eval Episodes

+2.05

PPO vs Greedy

Recap & Answers from Update 1

MDP Design (unchanged)

State

68-dim: user_embed(50) + genre_counts(18)

Action

18 genres → best movie by user history

Reward

Bernoulli($P(\text{like})$) — no manual fatigue

Episode

$T = 20$ steps

Window

10 recent genres tracked

Questions from Update 1

Q: One movie has multiple genres — how handled?

Movie-level $P(\text{like})$ used (not per-genre). genre_counts updated for ALL genres of the movie. Seen + rejected movies excluded from pool.

Q: Is RL actually useful for MovieLens?

Goal: verify RL learns better sequential strategy vs rule-based baselines — not to prove fatigue exists. PPO +2.05 vs Greedy confirms RL advantage even without fatigue signal.

Simulator Update: XGBoost → GRU4Rec

XGBoost (Update 1)

AUC = 0.734

- Static features: user_embed + movie_embed
- No sequence awareness
- Requires user history + SVD embeddings
- AUC higher on MovieLens (static data)

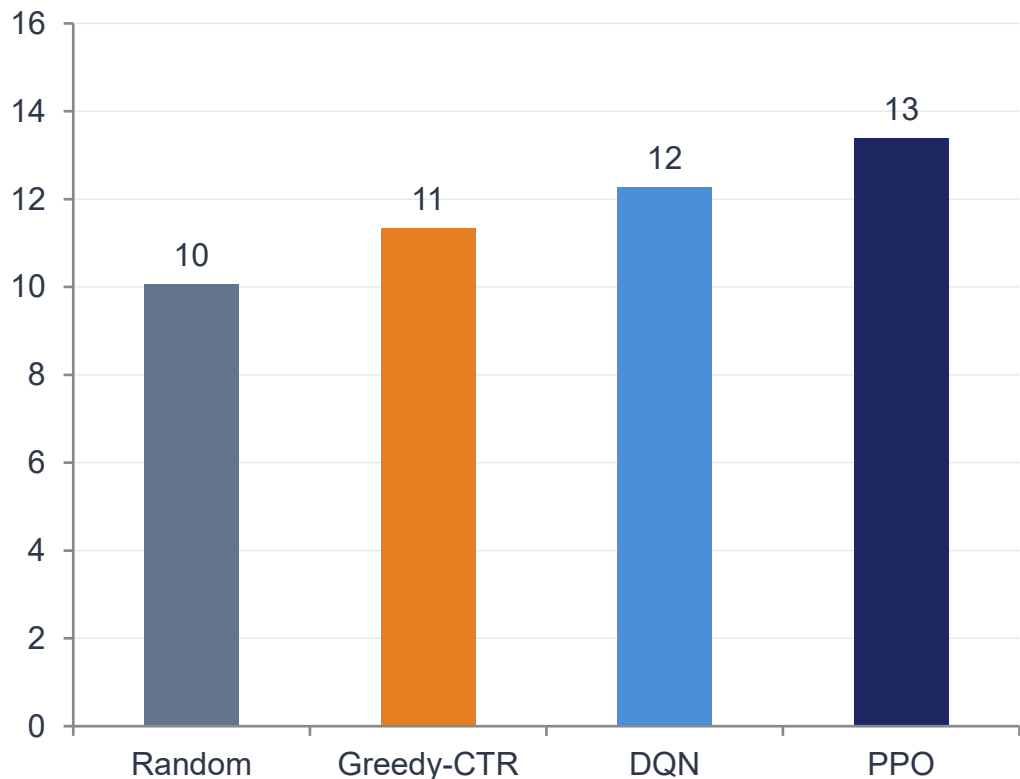
GRU4Rec (Update 2)

AUC = 0.703

- Sequence-aware: learns from viewing order
- Hidden state captures short-term dynamics
- Better suited for session-based datasets
- AUC lower — MovieLens lacks fatigue signal

Key Finding: GRU4Rec AUC < XGBoost — MovieLens lacks session-level fatigue structure (consistent with genre_counts ablation)

Main Results (T=20, 500 episodes)



+3.33

PPO vs Random

+2.05

PPO vs Greedy

+0.94

DQN vs Greedy

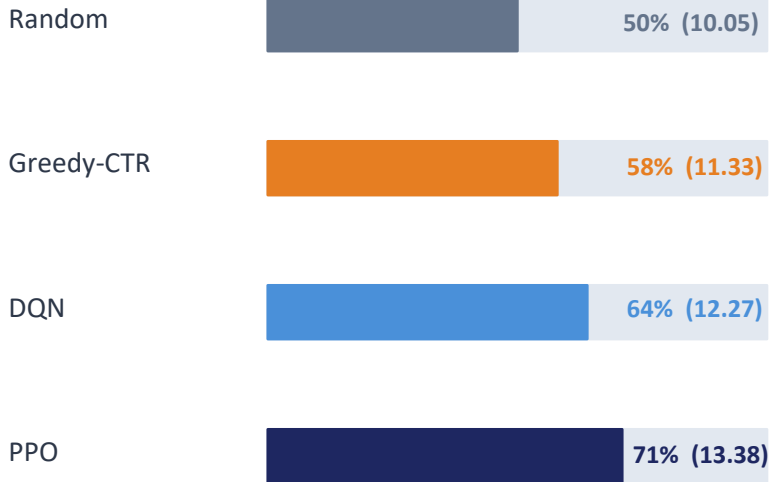
+1.11

PPO vs DQN

Interpreting the Results

Utilisation of Possible Range

$\text{Range} = [\text{clip_min} \times T, \text{clip_max} \times T] = [2.0, 18.0]$



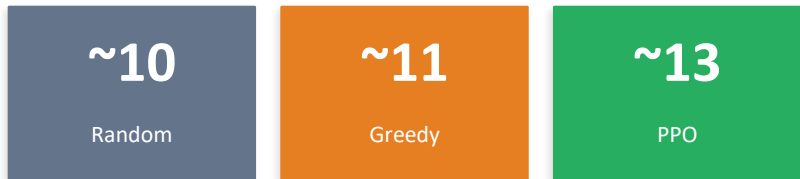
PPO utilises 71% of possible range vs 50% for Random

Practical Meaning (per session)

Per-step gain over Random:

$3.33 / 20 \text{ steps} = +0.167 p_like / \text{step}$

Liked movies per 20-step session:



Reduction in wasted recommendations:

PPO vs Random: 33% fewer unwanted movies

PPO vs Greedy-CTR: 24% fewer unwanted movies

Why PPO Outperforms DQN

DQN (Off-Policy)

Mean: 12.27

- Replay buffer mixes old + new data
- Early high- ϵ exploration data pollutes training
- Q-value estimates noisier
- Slower to adapt to short-horizon patterns

PPO (On-Policy)

Mean: 13.38

- Trains only on fresh interaction data
- $T=20$ short episodes favour on-policy updates
- Clipping prevents large destabilising updates
- Cleaner reward signal $\rightarrow$ better convergence

Conclusion: On-policy methods better suited for short-horizon ($T=20$) sequential recommendation

Bernoulli vs Expected Reward

Bernoulli Reward

$\text{reward} = \text{Binomial}(1, p_{\text{like}})$

Policy	Mean $\pm$ Std
Random	10.95 $\pm$ 2.20
Greedy-CTR	12.28 $\pm$ 2.43
DQN	12.44 $\pm$ 2.54
PPO	13.38 $\pm$ 2.79

PPO vs Greedy: +1.10

Sampling noise masks true policy advantage

P(like) Accumulation

$\text{reward} = \sum p_{\text{like}}$ (no sampling)

Policy	Mean $\pm$ Std
Random	10.98 $\pm$ 0.99
Greedy	12.16 $\pm$ 1.34
DQN	12.74 $\pm$ 1.40
PPO	13.79 $\pm$ 1.51

PPO vs Greedy: +1.63

True policy advantage revealed clearly

→ Expected reward is our primary evaluation metric going forward

Ablation Studies

① State: Remove genre_counts (68→50 dim)

DQN full (68-dim)	12.27 ± 3.12
DQN no genre (50-dim)	12.19 ± 3.42 $\Delta = -0.08$
PPO full (68-dim)	13.38 ± 2.79
PPO no genre (50-dim)	13.58 ± 2.78 $\Delta = +0.20$

Finding: genre_counts has minimal impact — consistent with MovieLens lacking fatigue signal. DQN slightly needs it; PPO does not.

② Window Size (PPO, T=20)

Window = 5	11.87 ± 2.87
Window = 10	11.80 ± 3.01 (current)
Window = 20	11.99 ± 2.73

Finding: Performance stable across window sizes ($\Delta < 0.2$). Model robust to history length choice. (Separately trained models — absolute values not directly comparable to main results)*

Limitations & Future Work

Limitations

MovieLens lacks fatigue signal

3 independent evidences: fatigue curve flat, GRU4Rec AUC < XGBoost, genre_counts ablation minimal

Cold-start problem

New users/movies have no SVD embedding — system cannot handle them without historical data

Offline evaluation only

All results from simulator — real-user validation absent; simulator bias may affect conclusions

Training variance

PPO/DQN results vary across runs without fixed seed — multi-seed evaluation needed

Future Work

Switch to KuaiRec / Tenrec

Session-based datasets with watch ratio & skip behavior — genuine fatigue signal for RL

Dynamic user embedding

Update SVD embedding at each step → richer state representation of current user mood

Cold-start solution

User: demographic group embedding. Movie: content-feature (genre/director) mapping

Multi-seed evaluation

Average over seeds [42, 123, 777] for statistically reliable PPO vs DQN comparison

Update 2 — Summary

- ✓ PPO (+2.05) and DQN (+0.94) both outperform Greedy-CTR — RL learns better sequential strategy
- ✓ PPO > DQN: on-policy learning better suited for short-horizon (T=20) recommendation
- ✓ Expected reward reveals true policy advantage masked by Bernoulli sampling variance
- ✓ Ablation confirms genre_counts minimally impactful — consistent with MovieLens lacking fatigue signal
- ✓ GRU4Rec AUC (0.703) < XGBoost (0.734) — further evidence MovieLens is not a fatigue-rich dataset
- Next: XGBoost vs GRU4Rec simulator ablation · Explore session-based dataset (KuaiRec)